NumPy Real-Life Problem

Sessions 1 & 2 · Arrays, Indexing, Slicing, Broadcasting

■ Scenario: Supermarket Sales Analysis

A local supermarket recorded daily sales (in ■) for **5 products** over **7 days** (one week). You are hired as a data analyst intern. Your job: clean the data, analyse it, and generate key business insights — all using NumPy.

Raw Sales Data (■) — Rows = Products, Cols = Days

Product	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Rice	1200	1350	1100	1400	1250	1800	1600
Oil	800	750	-50	820	790	1100	950
Biscuits	400	420	380	450	430	600	580
Milk	950	1000	920	980	1010	1200	1150
Soap	300	280	310	-20	295	400	370

■ Negative values (-50, -20) = data entry errors (bad data!)

Your Tasks

Task 1: Array Creation [Session 1]

Create the 5×7 NumPy array from the data above. Print its shape, ndim, size, dtype.

Your Code Here:

Task 2: Indexing & Slicing [Session 2]

Extract: (a) Rice sales only (b) Weekend sales (Sat+Sun) for all products (c) Oil sales on Wednesday (the bad value).

Your Code Here:

Task 3: Boolean Indexing [Session 2]

Find all negative (bad) values. Replace them with 0 using a boolean mask.

Your Code Here:

Task 4: Broadcasting [Session 2]

Apply a 5% weekend festival discount on ALL products for Sat & Sun using broadcasting.

Your Code Here:

Task 5: Aggregation [Session 1 — array methods]

Find: total weekly sales per product, best-selling day (column sum), average daily sale.

Your Code Here:

Concept → Task Map

Concept	Session	Used In
np.array(), shape, dtype, ndim	S1	Task 1
Row/column indexing	S2	Task 2
Boolean indexing (masking)	S2	Task 3
Broadcasting (scalar × array)	S2	Task 4
sum(), mean(), argmax()	S1	Task 5

■ Bonus Challenge

Which product had **zero sales on any day** after cleaning? Use boolean indexing + any() to find out. Then use broadcasting to add a flat ■50 delivery charge to all products for Mon & Tue only.